RAG Assignment Report

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Code Repository: https://github.com/Keerthana387/LMA_A1_RAG

1 Architecture Comparison

The main difference between a Bi-Encoder and a Cross-Encoder comes down to when and how the query and document tokens get to interact:

- **Bi-Encoder (`all-MiniLM-L6-v2`)**: In a Bi-Encoder, the query and document chunks are encoded completely separately. The model passes the query through the transformer to get one dense vector $\mathbf{u} = f(q)$, and does the same for the document chunk to get $\mathbf{v} = f(d)$. Their similarity is then computed using a quick vector operation like cosine similarity:

$$\text{sim}(q, d) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

  Because query and document never see each other during the transformer forward pass, there is no cross-attention between their tokens.
- **Cross-Encoder** (`ms-marco-MiniLM-L-6-v2`): In contrast, a Cross-Encoder takes both the query and the chunk concatenated together as one input sequence, typically formatted as `[CLS] query [SEP] chunk [SEP]`. They pass through the transformer together, allowing full all-to-all self-attention across all tokens at every layer. Every token in the query directly compares against every token in the document. This gives much higher semantic accuracy because the model can easily recognize exceptions, negative constraints, and contextual relevance.

2 Qualitative & Rank Shift Analysis

#	Category	Query	Target Rule	Rank Shift
1	Keyword Trap 1	Can players respond to a mana ability with an instant spell?	Rule 605.3b	#8 → #3
2	Exceptions	What happens if a blocker is removed before damage assignment?	Rule 510.1c	#8 → #1, #5 → #2
3	Negation / Timing	Can a player cast a creature spell when the stack is not empty?	Rule 302.1	#5 → #1
4	Buried Chunk	What happens to Equipment if equipped creature leaves battlefield?	Rule 701.3d / 704.5n	#9 → #1, #6 → #2
5	Keyword Trap 2	Does damage to a planeswalker reduce its loyalty counters?	Rule 306.8 / 120.3c	#4 → #1, #6 → #2
6	Complex Restriction	Can a creature with defender be declared as an attacking creature?	Rule 702.3b	#8 → #2

Table 1: Rank transitions of authoritative rule chunks from Bi-Encoder (Top-10) to Cross-Encoder (Top-3).

1. **FM1: Keyword Trap / Distractor (Responding to Mana Abilities)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, there is no mention of whether players can respond to a mana ability with an instant spell. Therefore, the answer is not in the context."*
   - **Answer 2 (Re-Ranked)**: *"Based on the provided context, no, players cannot respond to an activated mana ability. Rule 605.3b states that an activated mana ability doesn't go on the stack, 'so it can't be targeted, countered, or otherwise responded to. Rather, it resolves immediately after it is activated' [3]."*
   - **Baseline Failure Analysis**: The baseline **omitted an essential clause**. Because Rule 605.3b was buried at rank #8, the baseline Top-3 only contained general mana definitions, causing the LLM to state that the text lacked sufficient information.
2. **FM2: Exceptions & Corner Cases (Removed Blocker Combat Damage)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, a creature remains blocked even if all the creatures blocking it are removed from combat [2]."*
   - **Answer 2 (Re-Ranked)**: *"Based on the provided context, if no creatures are currently blocking an attacking creature because they were destroyed or removed from combat before damage is assigned, the attacking creature assigns no combat damage [1]. Additionally, a creature remains blocked even if all the creatures blocking it are removed from combat [3]."*
   - **Baseline Failure Analysis**: The baseline **omitted an essential clause**. It only answered the creature's blocked status from glossary chunks, but completely omitted what happens to combat damage because the actual damage exception rule (Rule 510.1c) sat outside the Top-3 at ranks #5 and #8.
3. **FM3: Negation & Constraints (Creature Spell Stack Timing)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, there is no mention of whether a player can cast a creature spell when the stack is not empty. The text does not contain the answer to this question."*
   - **Answer 2 (Re-Ranked)**: *"Based on the provided context, a player who has priority may cast a creature card from their hand during a main phase of their turn only 'when the stack is empty' [1]. Therefore, a player cannot cast a creature spell when the stack is not empty."*
   - **Baseline Failure Analysis**: The baseline **omitted an essential clause and rule**. Because Rule 302.1 (which enforces that creature spells can only be cast when the stack is empty) sat at rank #5, the baseline Top-3 only had general casting rules, leaving the LLM unable to answer.
4. **FM4: Buried Relevant Chunk (Equipment When Creature Leaves Battlefield)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, the answer is not in the text."*
   - **Answer 2 (Re-Ranked)**: *"Based on the provided context, if an Equipment ceases to be attached to an object (such as when the object leaves the battlefield), that counts as 'becoming unattached' [1]. The Equipment remains on the battlefield [2]."*
   - **Baseline Failure Analysis**: The baseline **omitted an essential clause**. The specific detachment definition (Rule 701.3d) and state-based action (Rule 704.5n) sat buried at ranks #9 and #6, while baseline Top-3 held generic equipment definitions, causing the LLM to report no answer.
5. **FM5: Keyword Trap 2 (Planeswalker Damage vs. Loyalty Counters)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, the answer is not mentioned."*
   - **Answer 2 (Re-Ranked)**: *"Yes. Damage dealt to a planeswalker results in that many loyalty counters being removed from it (rules 306.8 and 120.3c)."*
   - **Baseline Failure Analysis**: The baseline **omitted the governing rule**. Keyword overlap on "damage", "planeswalker", and "loyalty" pulled trample excess damage and loyalty ability rules into the Top-3, while the direct rule that damage removes loyalty counters was buried at ranks #4 and #6.

6. **FM6: Complex Interaction / Restriction (Defender Attacking)**
   - **Answer 1 (Baseline)**: *"Based on the provided context, there is no mention of whether a creature with the defender ability can be declared as an attacking creature. Therefore, the answer is not in the context."*
   - **Answer 2 (Re-Ranked)**: *"Based on the provided context, no. A creature with defender cannot attack (Rule 702.3b)."*
   - **Baseline Failure Analysis**: The baseline **omitted the essential restriction clause**. The Bi-Encoder retrieved general attack declaration structures into the Top-3, leaving Rule 702.3b buried at rank #8, so the baseline LLM missed the restriction entirely.

3 Latency Profiling

Query	**Retrieval** (s)	**Re-rank** (s)	**LLM P1** (s)	**LLM P2** (s)	**Total P1** (s)	**Total P2** (s)
Sanity Check	0.026	0.111	1.052	0.695	1.078	0.832
FM1: Mana Ability Response	0.024	0.101	0.993	0.885	1.017	1.011
FM2: Removed Blocker Damage	0.044	0.101	0.842	0.986	0.885	1.130
FM3: Creature Spell Stack Empty	0.038	0.108	1.133	0.941	1.171	1.087
FM4: Equipment Creature Leaves	0.024	0.124	2.068	0.898	2.092	1.046
FM5: Planeswalker Damage Loyalty	0.021	0.109	1.016	0.776	1.037	0.906
FM6: Defender Attacking	0.027	0.098	5.856	1.039	5.882	1.164
Average	**0.029**	**0.107**	**1.851**	**0.889**	**1.880**	**1.025**

Table 2: Empirical latency profiling across all benchmark queries (values directly from notebook execution).

4 Scaling Question

We cannot simply run a Cross-Encoder across the entire document collection from the start because of the computational cost and query latency:

- **Computational Complexity**: A Bi-Encoder encodes the query once ($O(1)$) and performs a dot product or Approximate Nearest Neighbor (ANN) search over pre-computed document embeddings in $O(N)$ or $O(\log N)$ time, which takes only a few milliseconds. In contrast, a Cross-Encoder must concatenate the query with every chunk and run a full transformer forward pass with $O(N \cdot L^2)$ self-attention complexity over token sequence length L. It cannot reuse any pre-computed document representations.
- **Empirical Latency**: In our notebook experiments, re-ranking 10 candidates took 0.107 seconds ($\approx$ 10.7 ms per candidate pair). For our document corpus of 3,090 chunks, running the Cross-Encoder over every chunk would take:

$$3{,}090 \times 0.0107\,\text{s} \approx \mathbf{33.1} \text{ seconds per query}$$

  On a standard production corpus of 100,000 chunks, that would scale to $\approx$ **17.8** minutes for a single query, which is completely unfeasible for interactive RAG systems.

This is why standard RAG pipelines use a two-stage approach: the fast Bi-Encoder retrieves the top 10 candidates in milliseconds, and the Cross-Encoder re-ranks only those 10 to provide high precision without the massive latency penalty.